VDX-6390

Vortex86DX 1GHz Evaluation Board

Quick Start-up Manual

(Revision 1.0A)

DMP Electronics Inc.
8F, No. 12, Wu-Quan 7th Rd,.
Wu-Gu Industrial Park.
Wu-Gu Xiang, Taipei #248
Taiwan
Tel：+886 2 2298 0770
Fax: +886 2 2290 2335
Email : soc@dmp.com.tw

Copyright

The information in this manual is subject to change without notice for continuous improvement in the product. All rights are reserved. The manufacturer assumes no responsibility for any inaccuracies that may be contained in this document. And makes no commitment to update or to keep current the information contained in this manual.

No part of this manual may be reproduced, copied, translated or transmitted, in whole or in part, in any form or by any means without the prior written permission of the DMP Electronics Inc.

?Copyright 2008 DMP Electronics Inc.
Manual No. IUM63900000-01 Ver.1.0A Nov., 2008

Trademarks Acknowledgment

Vortex86DX? is the registered trademark of DMP Electronics Inc.

Other brand names or product names appearing in this document are the properties and registered trademarks of their respective owners. All names mentioned herewith are served for identification purpose only.

Table of Content .

Chapter 1 Introduction ……1
1.1 Packing list ……1
1.2 Specifications ….....2
Chapter 2 Hardware Installation ……3
2.1 Board outline…… 3
2.2 Connectors and Jumpers summary ……....4
2.3 Adjust CPU and DRAM clock speed…….5
Chapter 3 Software installation ……..7
3.1 BIOS version ……7
3.2 IDE Mode ……..7
3.3 Windows XP installation note ……...8
3.4 Linux installation note ……..13

Chapter 1

Introduction

VDX-6390 is the official evaluation board for Vortex86DX CPU. It offers an easy way to evaluate the Vortex86DX and most of its build-in I/O functions. Users can also change the CPU speed by adjusting the jumper setting to get the desirable performance and power consumption of their application.

To get more detail, you can download the VDX-6390 schematic from ftp.dmp.com.tw, username:vxdx, password:gc301, the file folder called DM-116A

1.1 Packing List

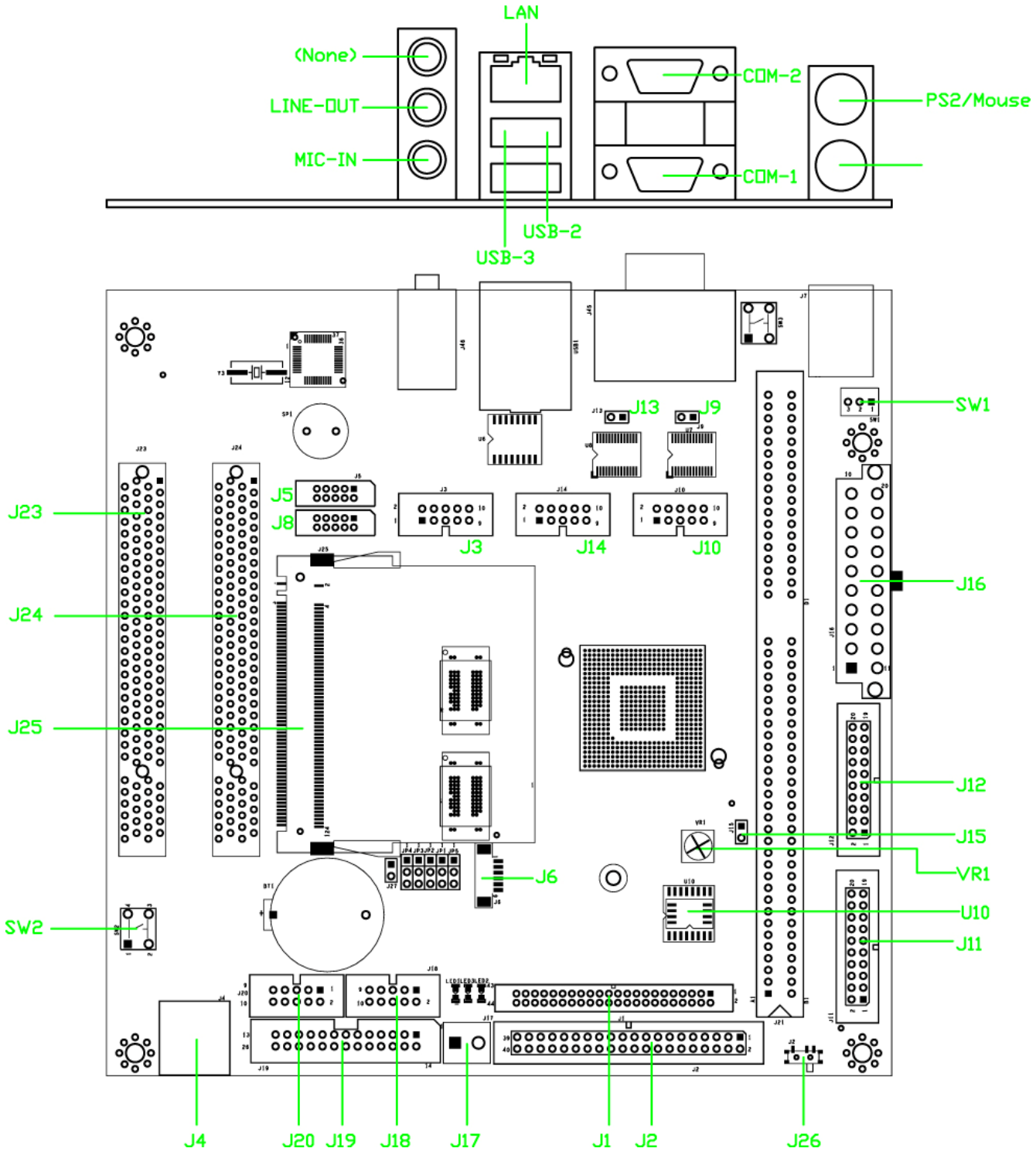
Product Name	Package
VDX-6390	● Vortex86DX CPU All-in-One Board ● RS232 Cable x 2 ● PRINTER cable x1 ● IDE Cable x 1 ● USB Cable x 1 ● GPIO Cable x 2

1.2 Specifications

Features	VDX-6390
CPU	DM&P Vortex86DX 600 MHz to 1GHz
BIOS	AMI BIOS – Core 8
System Memory	256MB DDR2 Onboard at 300MHz (DDR2-600)
Cache	L1:16K I-Cache, 16K D-Cache L2: 128KB
Bus Interface	ISA Bus x 2 PCI Bus x 1 MiniPCI Bus x 1
Watchdog Timer	2 Sets Software programmable from 30.5 us to 512 seconds.
LAN	Build-in R6040, 10/100Mbps Ethernet Controller
Audio chip	CM119 USB Audio Chip
I /O Interface	• Enhanced IDE port (UltraDMA-100/66/33) x2 • RS232 port x4 • Parallel port x1 • USB port x4 (USB 2.0 version) • GPIO ports • PWM channels • I2C bus • SPI bus • LPC bus • Redundancy bus • 10/100Mbps Ethernet port x1 • Audio, Line-out & Mic-in
Dimension	170mm X 170mm (6.69” x 6.69”)
Power Consumption	5V @ 600 mA (5V @ 880mA if with XGI Z9s MiniPCI VGA card and IDE Flash Disk)
Operating Temperature	-40°C ~ +75°C

Chapter 2
Hardware Installation

2.1 Board Outline



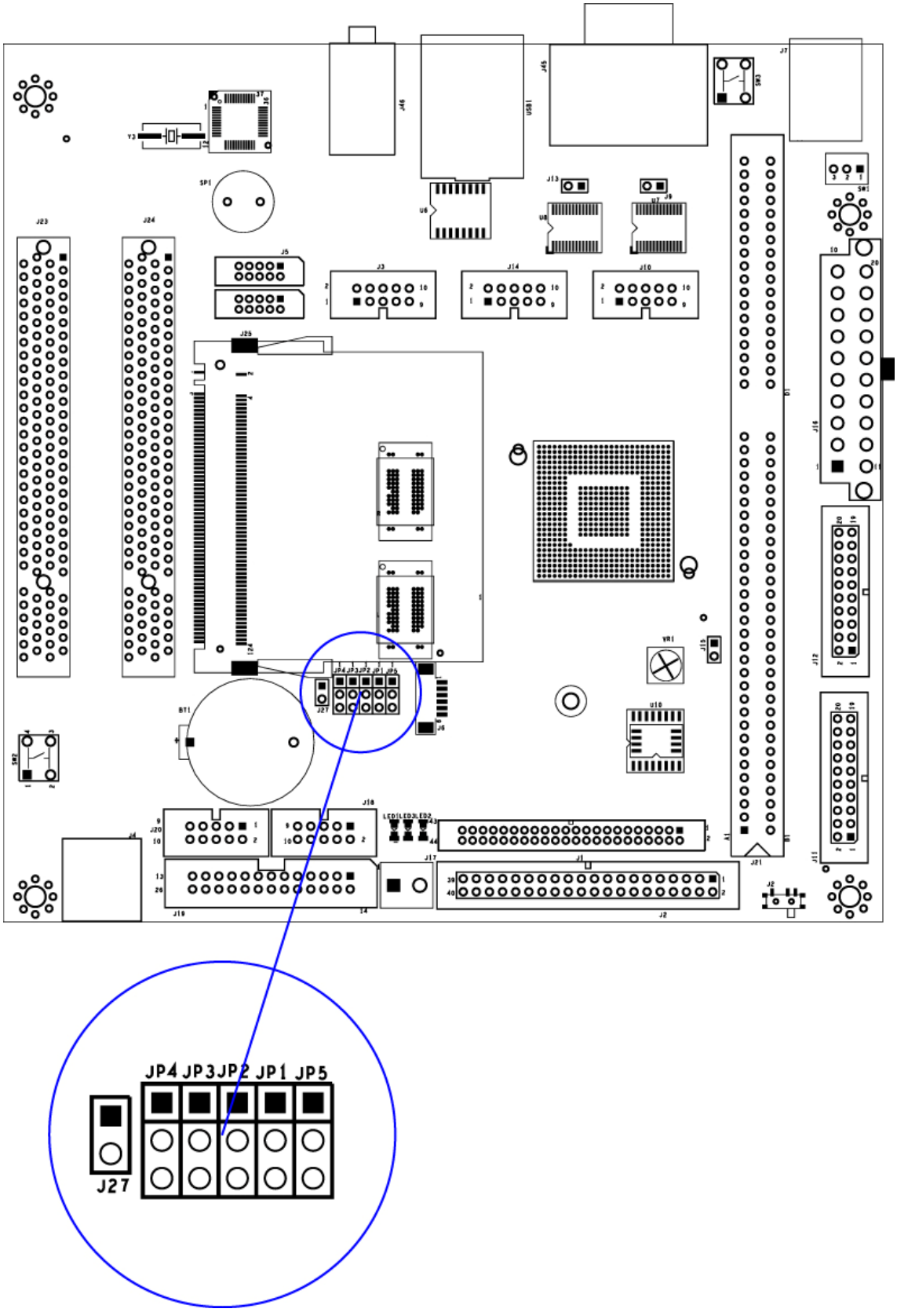
2.2 Connectors & Jumpers Summary

Below is the description for each port on VDX-6390, for the detail pin assignment, please refer to the schematic from ftp.dmp.com.tw, username:vxdx, password:gc301, file folder is DM-116A.

Nbr	Description	Note
J1	Primary IDE – 44 pin	
J2	Primary IDE – 40 pin	
J3	USB Port 0 & 1	The CM-119 Audio chip use USB Port 1, so the USB Port 1 is not functioned. If you don't want to use Audio, please remove R47 and R48, then populate the L4. You can refer to the schematic for detail.
J4	USB Client Port	USB Device port. Don't use, the driver is not ready yet
J5	LPC Bus	
J6	JTAG	You can close the jumper to disable JTAG
J7	PS2 KB/Mouse	
J8	Redundancy port	
J9	Disable COM-1	Close the jumper to disable COM-1, then enable J10
J10	COM1/P4	Adjust BIOS setting to use this port as TTL level COM1 or GPIO Port-4
J11	P0/P1/PWM	Adjust BIOS setting to use this port as GPIO Port-0,GPIO Port-1 or PWM channels
J12	P2/P3/SPI/I2C/PWM	Adjust BIOS setting to use this port as GPIO Port-2,GPIO Port-3 , SPI bus, I2C bus or PWM channels
J13	Disable COM-2	Close the jumper to disable COM-2, then enable J14
J14	COM2/ PWM Timer Counter	Adjust BIOS setting to use this port as TTL level COM2 or PWM Timer Counter
J15	Hold the U10 SPI Flash	Close the jumper to disable the U10 SPI Flash, then the SPI bus on J12 can work after adjusting the BIOS.
J16	ATX Power	
J17	DC-in +5V/GND	The VDX-6390 need +5V only, if you have the +5V DC/DC power adapter, you don't need to use the ATX Power.
J18	COM 3	COM 3, TTL level signal
J19	Printer port	Printer port I
J20	COM 4	COM 4, TTL level signal
J21	ISA Bus	8-bit/16-bit ISA bus
J23	PCI Bus	PCI -1
J24	PCI Bus	PCI -2
J25	MiniPCI	MiniPCI Socket
J26	Console switch	Switch "ON" to enable the Console Redirection automatically
27	JTAG Enable	Close the jumper to Disable the JTAG function

2.3 Adjust CPU and DRAM clock speed.

With VDX-6390, you can evaluate the different CPU and DRAM clock speed by adjust the JP1~JP5. Please see the detail as below.

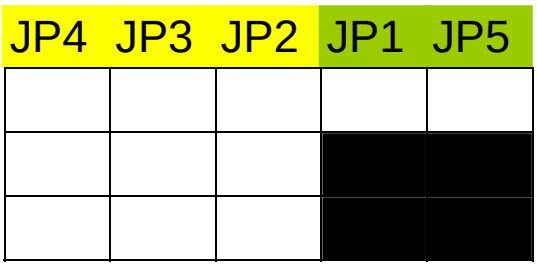


The tables as below show you how to change the CPU and DRAM speed.

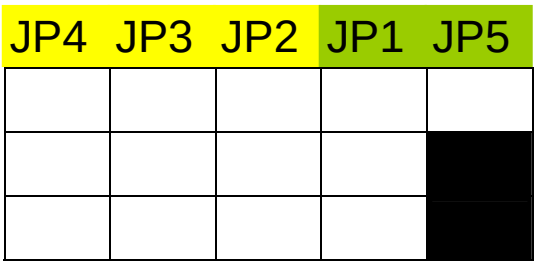
CPU	= 600 MHz
DRAM	= 200 MHz



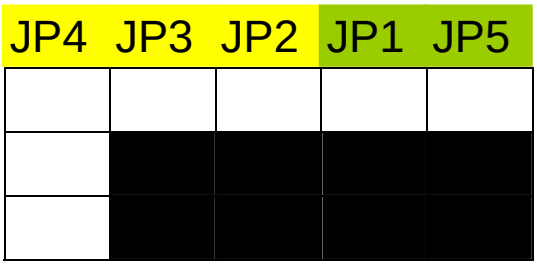
CPU	= 800 MHz
DRAM	= 266 MHz



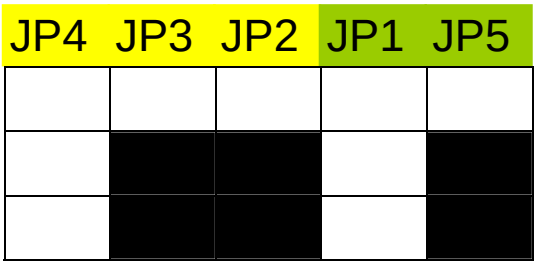
CPU	= 800 MHz
DRAM	= 300 MHz



CPU	= 1GHz
DRAM	= 266 MHz



CPU	= 1GHz
DRAM	= 300 MHz



Chapter 3

Software Installation

3.1 BIOS version

Please ensure the BIOS version on your VDX-6390 is A4 or later. You can always get the latest BIOS in a folder called BIOS from ftp.dmp.com.tw , username: vxdx, password:gc301



Be sure it's "BIOS Date:10/13/2008 VSX/VDX-6390 A4 Ver. Beta B" or later.

3.2 IDE mode

The current sample version of Vortex86DX has the incompatibility with specific operating systems. You have to install an external driver before you install the Windows XP and should be forced the IDE mode as PIO mode when you install the Linux. Please see below section 3.3 and 3.4 for detail.

To recognize the version of Vortex86DX by checking the printing code on the IC.
Sample version : 0837-B-PB
Mass production version: 09xx-C-PB

You don't have to fellow the 3.3 and 3.4 if the Vortex86DX on your VDX-6390 is 09xx-C-PB.

3.3 Windows XP installation note

To have the best performance with Windows XP, please be sure the IDE setting is correct in the BIOS. As you can see below.

Note: The installation time of Windows XP will take around 90 minutes. And the boot-up time of Windows XP is around 60 seconds. If you take longer time than that, it's means either the BIOS setting is wrong or your IDE Flash Disk (like CF or DOM) / HardDisk is too slow. 錯誤! 找不到索引項目。

The "Onboard IDE Operate Mode" is "Native Mode"



The "PCI IDE BusMaster" is "Enable"



To install the Windows XP, please fellow the steps as below.

How to install Windows XP on Vortex86DX

This document only applies to the Vortex86DX sample version A (code 0817-A-PB) and B (code 0837-B-PB). The mass production version C (code 09xx-C-PB , available date January 2009) will not need to load the IDE driver from USB FDD, just use the normal installation steps.

Due to the hardware incompatibility of the IDE interface on the sample version (code 0817-A-PB & code 0837-B-PB) of Vortex86DX, users should install the IDE driver manually when install the Windows XP. Below is the step by step guide.

After the installation, users should install the correct LAN and VGA driver.

---Required hardware---

- Vortex86DX CPU board with 256MB DDR2 DRAM or above.
- USB Floppy Drive or 1.44" Floppy Drive.
- USB Keyboard / Mouse (or PS/2 Keyboard / Mouse)
- IDE CABLE
- CD-ROM
- Compact Flash 2GB (or above), IDE Flash Disk or Hard Disk.

---Required software---

- Windows XP Service Pack 1/2/3 installation CD. (English version is preferred)
- IDE Driver for Vortex86DX.

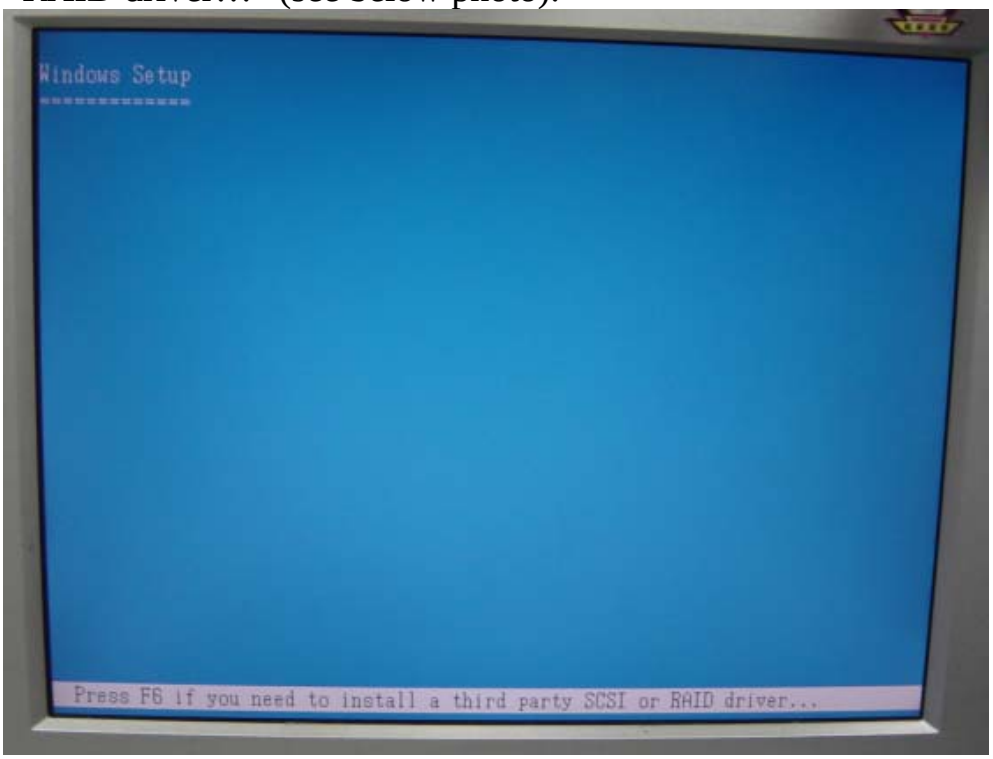
---STEPS--

1, Copy the folder of attached driver called “Windows” and " txtsetup.oem " to USB Floppy Drive or 1.44" Floppy Drive, and before the format of FDD disk is FAT12.

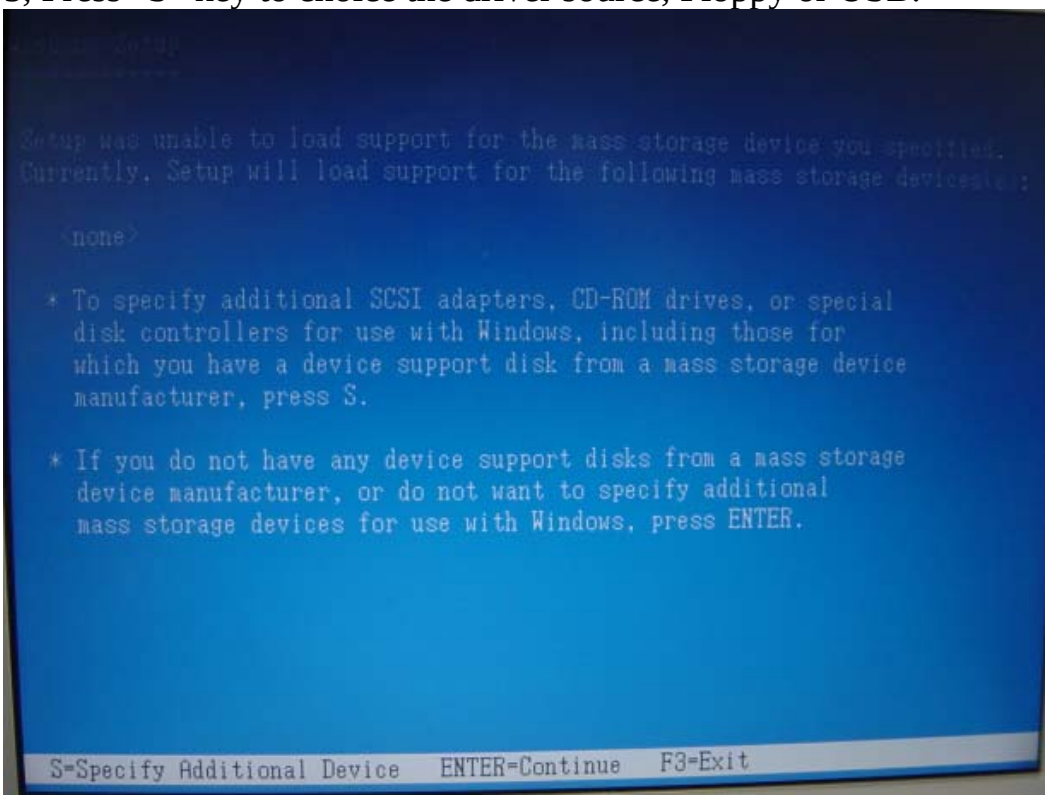
2, Make sure your platform is connected with above USB Floppy Drive (or 1.44" Floppy Drive) , Keyboard / Mouse, CD ROM (with XP CD) and Compact Flash (or Hard disk).

3, Power-on the platform and press the F11 to choose CD-ROM booting selection. Also, make sure the BIOS released date is after 2008/10/13.

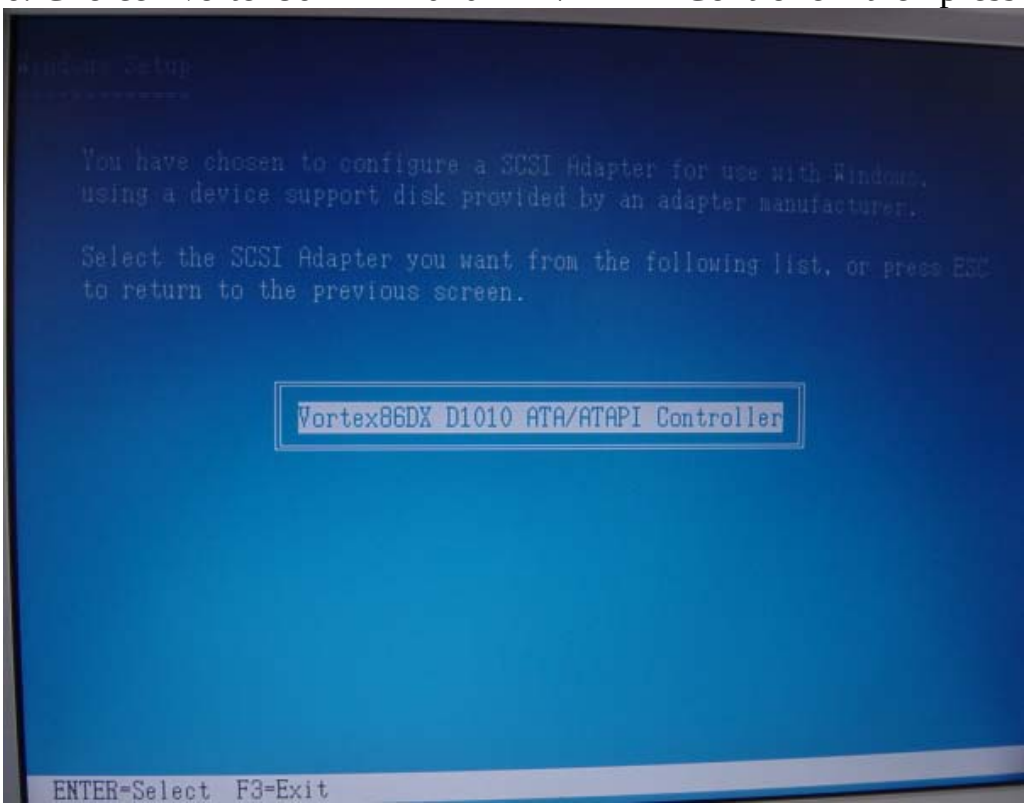
4, Press F6 when you see the “Press F6 if you need to install a third party SCSI or RAID driver…” (see below photo).



5, Press “S” key to choice the driver source, Floppy or USB.

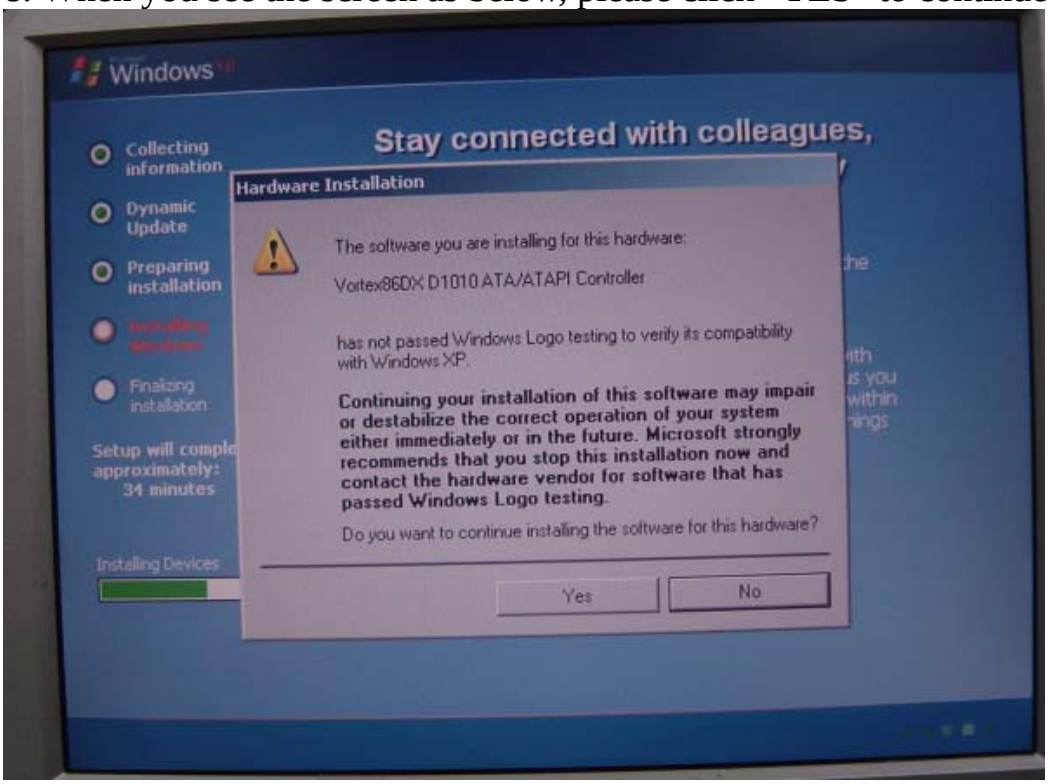


6. Choice “Vortex86DX D1010 ATA/ATAPI Controller” then press the “Enter Key”.



7. And then, please follow the installation steps.

8. When you see the screen as below, please click “YES” to continue.



9. Continue the installation.

10. After the first time boot up, you should install the correct LAN and VGA driver.

- If you are using Vortex86DX build-in LAN controller, please use attached LAN dirver.
- If you are using XGI Z9s VGA chip, please use attached VGA driver.

11. Done.

3.4 Linux installation note

The current sample version ((code 0817-A-PB & code 0837-B-PB) of Vortex86DX has the incompatibility with Linux.

To temporarily solve the problem until the mass production (code 09xx-C-PB) version is available. Please forced the IDE setting in BIOS to PIO mode before install Linux on Vortex86DX.

- Forced the IDE on PIO Mode. (simple way, but the IDE speed will be slow).

The “Onboard IDE Operate Mode” is “Legacy Mode”



The “PCI IDE BusMaster” is “Disable”



---------- END ----------